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Traditio et Innovatio

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Time-Domain Radiation Model of a Cap on a Rigid Sphere

Introduction

The velocity-to-pressure exterior wave field expansion for spherically shaped radiators is well known [1, 2]. It involves a modal radial filter—based on the spherical Hankel function and its derivative—which can be given as a Laplace-domain transfer function. Its frequency-dependent part can be expressed as a partial fraction expansion and is consequently suited for the impulse invariance digital filter design. This was utilised for radial directivity control of spherical loudspeaker arrays [3], which essentially deals with the inverse filters compared to the radiation case discussed here.

Recently, a band-limited impulse invariance method (BLIIM) was introduced [4]. This improves the diffraction filters for spherical microphone array models by reduction of time-domain aliasing [5]. Here, we apply BLIIM for the radiation model. For a given velocity profile of spherical cap pistons on an otherwise rigid sphere, the direction-dependent radiation and the far-field directivity of such sources can be modelled with dedicated digital filters. We discuss a numerical implementation for a dodecahedron source, which could be utilised within an image source model and as theoretical time-domain model in the field of directional (room) impulse responses.

Radiation Model

Exterior wave field expansion with radial filter frequency response $H_n(\omega, r, R)$ using the spherical Hankel function and its derivative, cf. [1, 2]

$$p(\omega, r, \boldsymbol{\theta}) = \rho_0 c \sum_{n=0}^{N} \overbrace{-\mathrm{j}\frac{h_n^{(2)}(\frac{\omega}{c}r)}{h_n^{\prime(2)}(\frac{\omega}{c}R)}}^{H_n(\omega,r,R)} \cdot \overbrace{\sum_{m=-n}^{+n} \underbrace{\sum_{l=1}^{L} \sqrt{\frac{4\pi}{2n+1}} g_l\, v_{n0}\, Y_{nm}^*(\boldsymbol{\theta}_l)}_{v_{nm}}\, Y_{nm}(\boldsymbol{\theta})}^{v_n(\boldsymbol{\theta})}$$

Expansion with radial filters as Laplace-domain transfer function ($\mathrm{j}\omega \to s$)

$$p(s, r, \boldsymbol{\theta}) = \rho_0 c\; R\, \frac{\mathrm{e}^{-s\cdot(\frac{r-R}{c})}}{r} \sum_{n=0}^{N} \Big(1 + \underbrace{\hat{H}_n(s, r, R)}_{n\,\text{zeros},\, n+1\,\text{poles}}\Big) \cdot v_n(\boldsymbol{\theta})$$

Variables

- $\boldsymbol{\theta} = [\text{polar angle}\,\theta,\ \text{azimuth angle}\,\phi]$, distance r from spherical source's origin
- l-th cap piston facing to $\boldsymbol{\theta}_l$, constant velocity with gain g_l, source radius R with $r > R$
- unit-gain velocity of axial-symmetric north-pole cap with half-opening angle α:
  $v_{n0} = \sqrt{\frac{\pi}{2n+1}}\big(P_{n-1}(\cos\alpha) - P_{n+1}(\cos\alpha)\big)$ [6]
- spherical harmonics $Y_{nm}(\cdot)$, Legendre polynomial $P_n(\cdot)$ following [6]
- $\mathrm{e}^{+\mathrm{j}\omega t}$ convention, $\mathrm{j}^2 := -1$, angular frequency ω, time t, sampling frequency f_s
- speed of sound c, density of air ρ_0, sound pressure p, velocity v
- sound propagation delay $\tau = \frac{r-R}{c}$ should be modelled with a suitable fractional delay filter

Radial Filter

Partial fraction expansion for $\hat{H}_n(s, r, R)$ yields a parallel, continuous-time IIR filter bank [5]. The additional pass-thru, unit-gain system is also parallel.

$$1 + \hat{H}_n(s, r, R) = \begin{cases} = \dfrac{s\cdot \sum_{k=0}^{n} \beta_n(k)\cdot(\frac{r}{c})^{k-n}\cdot s^k}{\sum_{k=0}^{n+1} \gamma_n(k)\cdot(\frac{R}{c})^{k-n-1}\cdot s^k} = 1 + \sum_{k=0}^{n} \dfrac{\mathrm{Res}_{n,k}}{s - \frac{c}{R}\mathring{s}_{\gamma_n(k)}} & \text{, general} \\[2ex] \approx \dfrac{s\cdot s^n}{\sum_{k=0}^{n+1} \gamma_n(k)\cdot(\frac{R}{c})^{k-n-1}\cdot s^k} = 1 + \sum_{k=0}^{n} \dfrac{\mathrm{Res}^{\mathrm{Far}}_{n,k}}{s - \frac{c}{R}\mathring{s}_{\gamma_n(k)}} & \text{, for } h_n^{(2)}\big(\frac{\omega}{c}r \gg 1\big) \end{cases}$$

For the n-th order denominator polynomial, $\gamma_n(k)$ coefficients yield $n+1$ roots (i.e. the $n+1$ normalised poles of the radial filter) with the k-th root $\mathring{s}_{\gamma_n(k)}$ and the k-th residual $\mathrm{Res}_{n,k}$ or $\mathrm{Res}^{\mathrm{Far}}_{n,k}$, respectively, cf. [5].

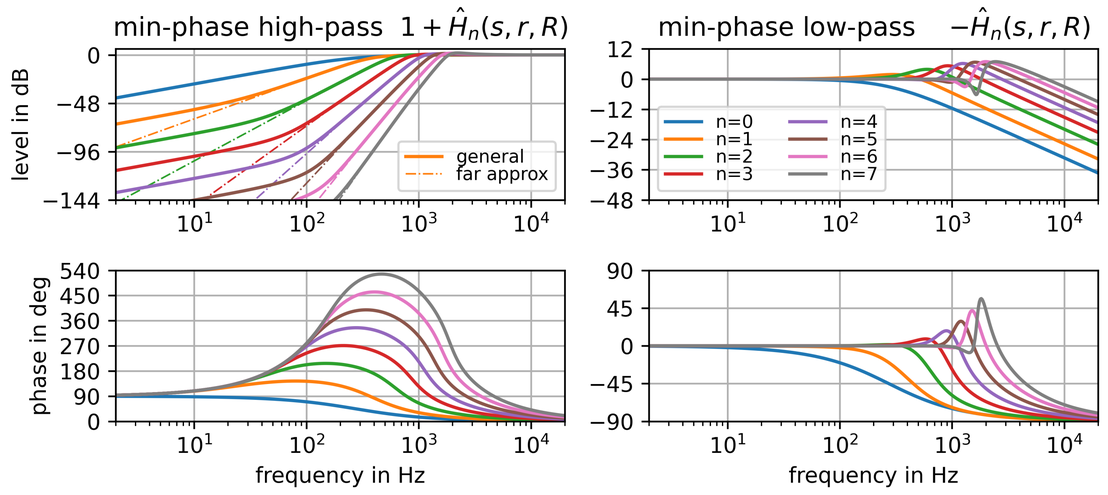
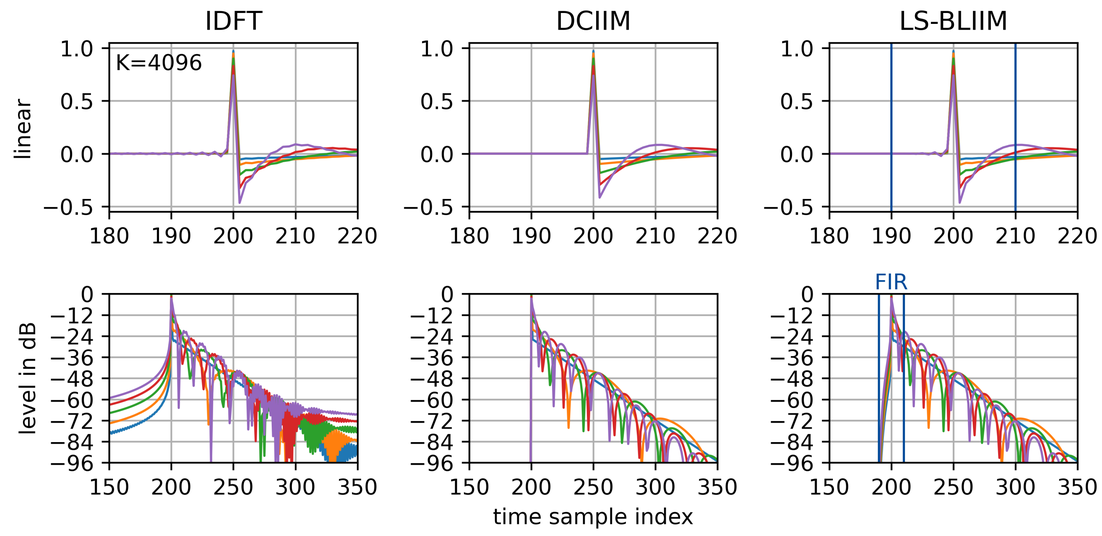


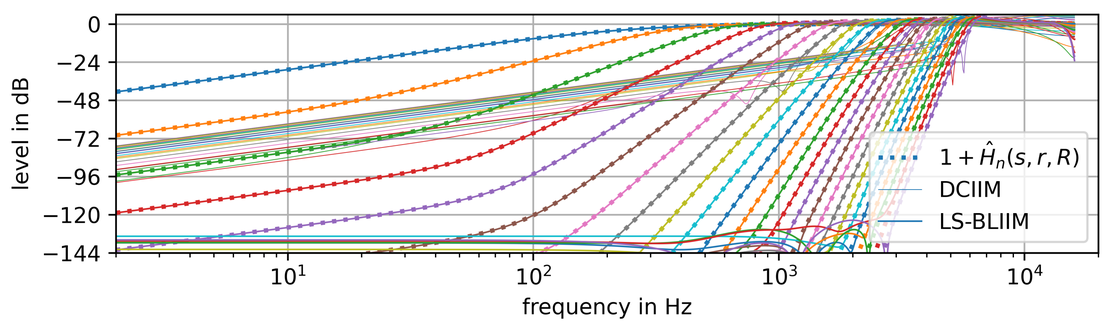
Figure 1: Radial filters for $r = 2$ m, $R = 0.2$ m, orders $n = 0\ldots7$, $c = 343$ m/s.

Band-Limited Impulse Invariance Filter Design

BLIIM uses a short, parallel, anti-aliasing-acting FIR filter to smooth the transient of the impulse [4] . A least-squares (LS) design of the FIR is utilised [5]. As a reasonable trade-off-strategy, this so called LS-BLIIM outperforms IDFT in the time domain and non-band-limited IIM in the frequency domain.



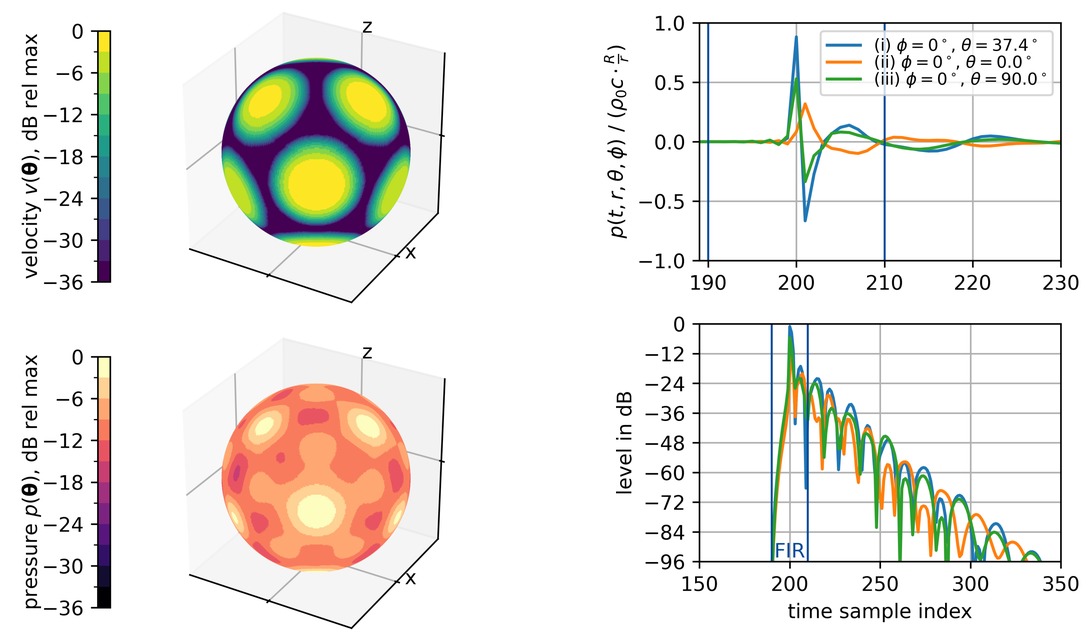
(a) Radial filter impulse responses linear (top) and dB (bottom) for orders $n = 0\ldots4$. Frequency-sampling / IDFT (left), DC-matched IIM (middle), least-squares band-limited IIM (right). Delay τ is included in the filters.



(b) Radial filter frequency responses for orders $n = 0\ldots24$. DCIIM is less accurate than LS-BLIIM; both are increasingly inaccurate for higher orders and/or closer to $f_\mathrm{s}/2$ depending on the R/r-ratio.

Figure 2: LS-BLIIM digital-filter design for $\big(1 + \hat{H}_n(s, r, R)\big)$ following [5], $f_\mathrm{s} = 32$ kHz, $c = 343$ m/s, $R = 0.2$ m, $r = 200 \cdot \frac{c}{f_\mathrm{s}} + R = 2.34375$ m, LS-BLIIM parameters: 21 FIR coefficients, 27 log-spaced control frequencies $[62\ldots16\mathrm{k}]$ Hz.

Case Study: Dodecahedron Source



(a) Velocity $v(\boldsymbol{\theta})$ at R (top), sound pressure $p(\boldsymbol{\theta})$ in the 4 kHz third-octave band at r (bottom).

(b) Pressure impulse responses in the direction of a cap, the north pole and the x-axis, respectively.

Figure 3: $N = 24$, $f_\mathrm{s} = 32$ kHz, $c = 343$ m/s, $R = 0.2$ m, $r = 200 \cdot \frac{c}{f_\mathrm{s}} + R = 2.34375$ m, 12 caps on the faces of a regular dodecahedron, $\alpha = 20$ deg, caps level with ±2 dB uniform-PDF dither, spatially low-pass filtered spectrum v_n with $\beta_{\text{Kaiser Window}} = 5$.

References

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[4] N. Hahn, F. Schultz, S. Spors (2022): Band Limited Impulse Invariance Method. 30th EUSIPCO, Belgrade.

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Code @ `https://github.com/spatialaudio/daga2026_dode`